Database Technologies – Assignment 9

Create appropriate indexes to solve the following problems.

Note: use "sales" database to solve below queries.

1. Create an index that will enable a user to pull orders for '1990-10-03' out of the Orders table quickly.

2. If the Orders table has already been created, how can you force the onum field to be unique (assume all current values are unique)?

3. Create an index that would permit salesperson to retrieve his orders.

4. Let us assume that each salesperson is to have only one customer of a given rating, and this is currently the case. Create an index that enforces it.

5. Create an index to speed up searching order on a given date by given customer.

SUNBEAM

Sunbeam Institute of Information Technology Pune & Karad

1.

```
d6_96728_shaurya> CREATE INDEX idx_odate
    -> ON orders(odate);
Query OK, 0 rows affected (0.646 sec)
Records: 0  Duplicates: 0  Warnings: 0

d6_96728_shaurya> SHOW INDEX FROM orders;
+--------+------------+-----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| Table  | Non_unique | Key_name  | Seq_in_index | Column_name | Collation | Cardinality | Sub_part | Packed | Null | Index_type | Comment | Index_comment | Visible | Expression |
+--------+------------+-----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
| orders |          1 | idx_odate |            1 | odate       | A         |           2 |     NULL |   NULL | YES  | BTREE      |         |               | YES     | NULL       |
+--------+------------+-----------+--------------+-------------+-----------+-------------+----------+--------+------+------------+---------+---------------+---------+------------+
1 row in set (0.114 sec)

d6_96728_shaurya> |
```

2.

```
d6_96728_shaurya> CREATE UNIQUE INDEX idx_onum
    -> ON orders(onum);
Query OK, 0 rows affected (0.310 sec)
Records: 0  Duplicates: 0  Warnings: 0

d6_96728_shaurya>
```

3.

```
d6_96728_shaurya> CREATE INDEX idx_snum
    -> ON orders(snum);
Query OK, 0 rows affected (0.479 sec)
Records: 0  Duplicates: 0  Warnings: 0

d6_96728_shaurya>
```

4.

```
ERROR 1064 (42000): You have an error in your SQL syntax; check the manual that corresponds to your MySQL server version for the right syntax to use near 'WHERE odate = ? AND cnum = ?' at line 1
d6_96728_shaurya> SELECT snum, rating, COUNT(*)
    -> FROM customers
    -> GROUP BY snum, rating
    -> HAVING COUNT(*) > 1;
+------+--------+----------+
| snum | rating | COUNT(*) |
+------+--------+----------+
| 1001 |    100 |        2 |
+------+--------+----------+
1 row in set (0.007 sec)

d6_96728_shaurya> SELECT *
    -> FROM customers
    -> WHERE snum = 1001 AND rating = 100;
+------+---------+--------+--------+------+
| cnum | cname   | city   | rating | snum |
+------+---------+--------+--------+------+
| 2001 | Hoffman | London |    100 | 1001 |
| 2006 | Clemens | London |    100 | 1001 |
+------+---------+--------+--------+------+
2 rows in set (0.006 sec)

d6_96728_shaurya> UPDATE customers
    -> SET rating = 200
    -> WHERE cnum = 2006;
Query OK, 1 row affected (0.124 sec)
Rows matched: 1  Changed: 1  Warnings: 0
```

5.

```
d6_96728_shaurya> SELECT *
    -> FROM orders
    -> WHERE odate = '1990-10-03'
    -> AND cnum = 2008;
+------+---------+------------+------+------+
| onum | amt     | odate      | cnum | snum |
+------+---------+------------+------+------+
| 3001 |   18.69 | 1990-10-03 | 2008 | 1007 |
| 3006 | 1098.16 | 1990-10-03 | 2008 | 1007 |
+------+---------+------------+------+------+
2 rows in set (0.041 sec)
```